

GARVIT GUPTA

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EDUCATION

B.Sc. (Hons.) Mathematics with Research, DDU College, Delhi University

Jul 2022 – Jun 2026

CGPA: 8.0 / 10 · Relevant coursework: Statistical Modeling, Linear Algebra, Optimization Theory, Probability & Stochastic Processes.

EXPERIENCE

ML & Mathematics Researcher — Final Year Thesis — DDU College, Delhi University

Aug 2025 – Jun 2026

- Designed a bias-mitigation algorithm combining statistical optimization with fairness-aware loss functions, reducing demographic bias up to 38% across 3 fairness metric families while holding accuracy within 1.2% of baseline.
- Built an EU AI Act-aligned Responsible AI platform (IBM AIF360, PyTorch, FastAPI, DagsHub) with a 5-phase automated bias-audit pipeline, cutting audit time from days to under 10 minutes across 20+ MLflow-tracked runs.

Machine Learning Intern — FastHire Manpower Solutions

Feb 2024 – Mar 2024

- Built an income classifier (Random Forest, SMOTE, GridSearchCV) on imbalanced socioeconomic data, reaching 91% accuracy and a 14% F1 improvement over the client's existing rule-based system.
- Shipped the model through a DagsHub CI/CD pipeline with automated experiment tracking, cutting the release cycle from 2 weeks to same-day.

PROJECTS

SCOUT — Supply Chain Disruption Early-Warning Agent [Live Demo ↗](#) · [GitHub ↗](#)

LangGraph · Gemini · XGBoost · MLflow/DagsHub · DVC · FastAPI · MySQL (Aiven) · HTML/CSS/JS

- Built a 3-agent LangGraph pipeline (Monitor → Risk → Recommend) that scores shipment disruption risk and delay using two XGBoost models (ROC-AUC 0.76) trained on 180K+ real shipment records.
- Deployed the pipeline as a live FastAPI service with Gemini-generated recommendations and a custom HTML/CSS/JS dashboard on GitHub Pages, tracked end-to-end through MLflow and a DVC pipeline.

Claim-Sentinel — Insurance Fraud Detection & Auto-Adjudication Agent [Live Demo ↗](#) · [GitHub ↗](#)

LangGraph · LightGBM · SHAP · Groq LLM · FastAPI · MySQL (Aiven) · HTML/CSS/JS

- Designed a 3-agent LangGraph pipeline (Intake → FraudRisk → Adjudication) around a class-balanced LightGBM classifier (ROC-AUC 0.83) with SHAP-based explainability for every prediction.
- Deployed the system on Render with cloud MySQL and a custom HTML/CSS/JS interface, auto-approving low-risk claims and escalating high-risk ones with LLM-generated, audit-logged adjuster notes.

Reverse-Engineering the Negation Circuit in GPT-2 (Mechanistic Interpretability) [GitHub ↗](#)

PyTorch · Activation Patching · Direct Logit Attribution · Causal Ablation

- Used activation patching and direct logit attribution — two independent causal methods — across 960 minimal-pair prompts to localize the attention head driving GPT-2's sentiment shift under negation.
- Ran ablation studies showing a distributed, redundant circuit: removing the top 10 implicated heads cut 37% of the effect, versus ~1% for the single strongest head.

AURUM — Gold Price Direction Intelligence [GitHub ↗](#)

FastAPI · Scikit-learn · MLflow · DagsHub Model Registry · Walk-Forward CV · Render.com · HTML/CSS/JS + Chart.js

- Trained an ElasticNet-regularized logistic regression classifier on 70 years of gold-price data using walk-forward cross-validation, reaching 83% directional accuracy across 50+ MLflow-tracked experiments.
- Deployed the model as a live FastAPI service on Render with a custom HTML/CSS/JS + Chart.js frontend showing 7-day predictions, live INR conversion, and confidence-scored trend charts.

TECHNICAL SKILLS

Languages Python · SQL · R · LaTeX · Mathematica · HTML/CSS/JS

ML / DL PyTorch · TensorFlow/Keras · Scikit-learn · XGBoost · LightGBM

LLMs / Agents LangGraph · LangChain · Gemini & Groq APIs · Transformer Fine-tuning · Prompt Engineering

XAI SHAP · LIME · Activation Patching · Direct Logit Attribution · Bias Detection & Mitigation

MLOps MLflow · DagsHub · DVC · GitHub Actions CI/CD · Docker · Apache Airflow

Deployment FastAPI · MySQL (Aiven) · MongoDB · Streamlit · Cloud (AWS · OCI · Render)

CERTIFICATIONS

Claude Code in Action · Anthropic · Mar 2026 · [Verify ↗](#)

Oracle Cloud Infrastructure 2025 Certified Generative AI Professional · Oracle University · Oct 2025

Oracle Cloud Infrastructure 2025 Certified Data Science Professional · Oracle University · Sep 2025